Global Trade Explorer — Dependencies, Flows & Disruptions

COM-480 Data Visualization · Milestone 2

Team COM-480 Trade Viz

2026-04-17

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Dependencies, Flows & Disruptions

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> An interactive dashboard that makes global trade readable: *who trades what with whom*, **how concentrated** each country's trade is, **how exposed** it is to a single partner, and **how shocks propagate** through the network — across 2000–2023 and five commodity families.

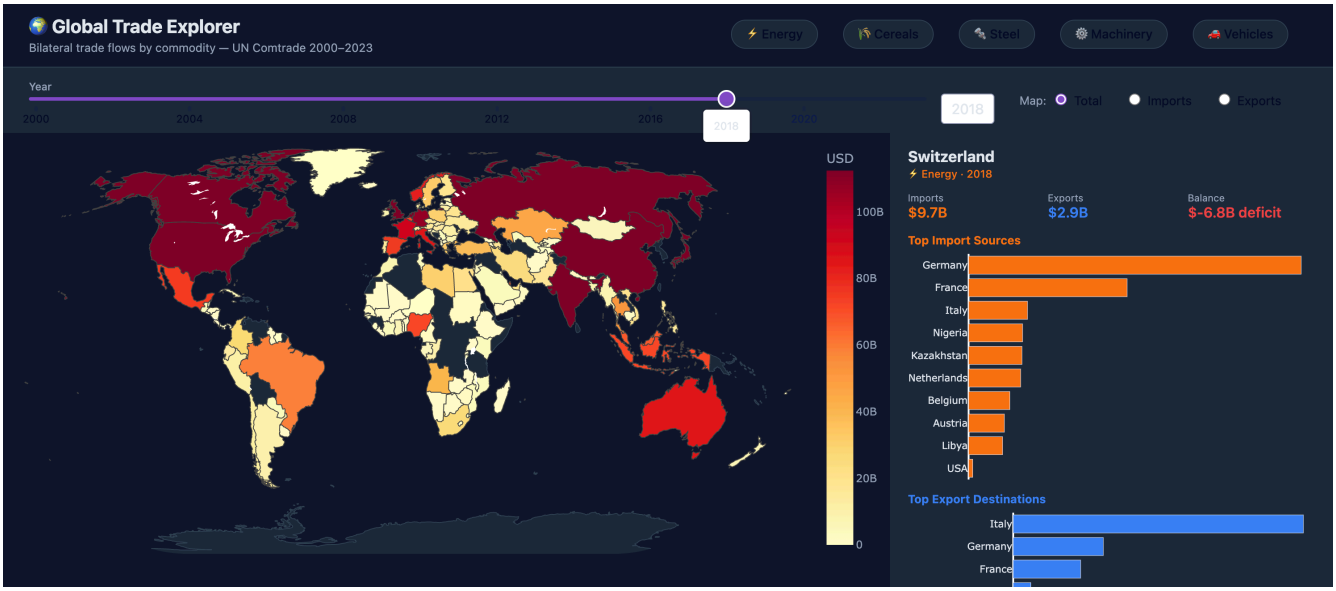

Figure 1: Hero: global energy-trade choropleth with year slider and country-detail panel (prototype)

Live prototype: https://global-trade-viz.onrender.com **Repository:** https://github.com/com-480-data-visualization/Vizion

1. Project Goal & Audience

We are building an interactive web dashboard that lets users explore global bilateral trade flows (UN Comtrade, 2000–2023) across several commodity families — **energy (HS 27)**, **cereals (HS 10)**, **iron & steel (HS 72)**, **machinery (HS 84)** and **vehicles (HS 87)**.

Existing platforms (OEC, World Bank, UN Comtrade dashboards) answer *what* was traded but rarely expose *structure* — concentration, dependency, and fragility. Our dashboard pairs a familiar entry point (a choropleth world map) with progressively deeper analytical views: time series with shock annotations, Sankey flows, a per-country dependency panel, a "top movers" bar, an animated dependency-race, and — as a stretch goal — a **disruption simulator** that removes a country/link and shows how supply shortfalls propagate.

Target audience. Students, researchers, and citizens interested in global economics, geopolitics and supply-chain resilience. The design prioritizes *progressive disclosure*: a newcomer sees a map; a power user can drill into HHI values, animate dependency over time, and compare two countries side-by-side.

2. Dataset

UN Comtrade, annual bilateral flows (reporter × partner × HS code × flow × USD), 2000–2023. After cleaning (year filter, ISO-3 normalization, removal of aggregate entities such as "World" and "Areas, nes") we ship three processed commodities (energy 393 k rows, cereals 230 k, steel 329 k) and are downloading two more (machinery, vehicles) under the Comtrade free-tier rate limit. Limitations: re-export inflation for hub countries (NLD, BEL, SGP), nominal USD (no inflation adjustment), sparse coverage pre-2005.

3. Visualizations

All views below are either **already working** in the prototype or scaffolded behind a tab. Screenshots are from the running dashboard.

(A) Choropleth World Map — entry point

Log-scaled USD, year slider (2000–2023), commodity dropdown (Energy / Cereals / Steel / Machinery / Vehicles), import/export/total toggle. Click a country opens the detail panel (B–E). See the hero screenshot on page 1 (Switzerland selected, Energy 2018 — import sources dominated by Germany and France, export destinations led by Italy).

(B) Trade History — country detail

Dual line (imports vs exports) over 2000–2023 with shock annotations (2008 crisis, 2014 oil crash, 2020 COVID, 2022 shock).

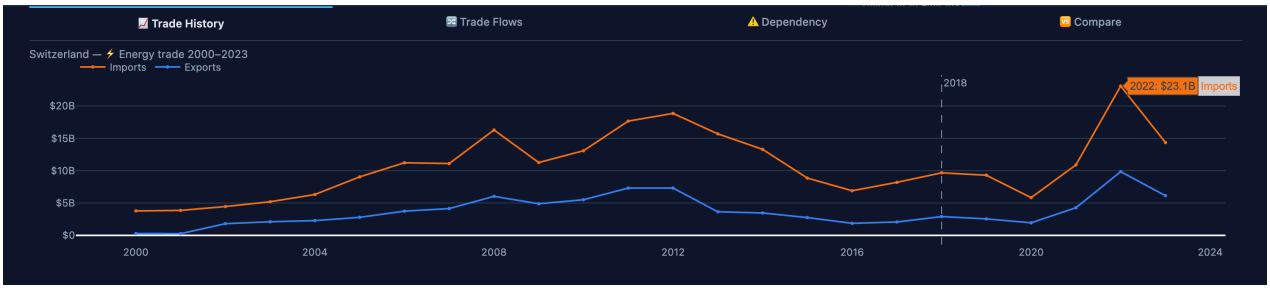

Figure 2: Country detail panel — Trade History tab

(C) Sankey Flow Diagram — bilateral structure

Left = import sources, right = export destinations, ribbon width = USD value for the selected country, commodity and year.

Figure 3: Trade Flows tab — Sankey of Switzerland's energy trade, 2018

(D) Dependency Panel — concentration over time

Top-partner share and top-3 partners share traced across 2000–2023, with "Diversified / Moderate / Concentrated" bands to contextualize the value.

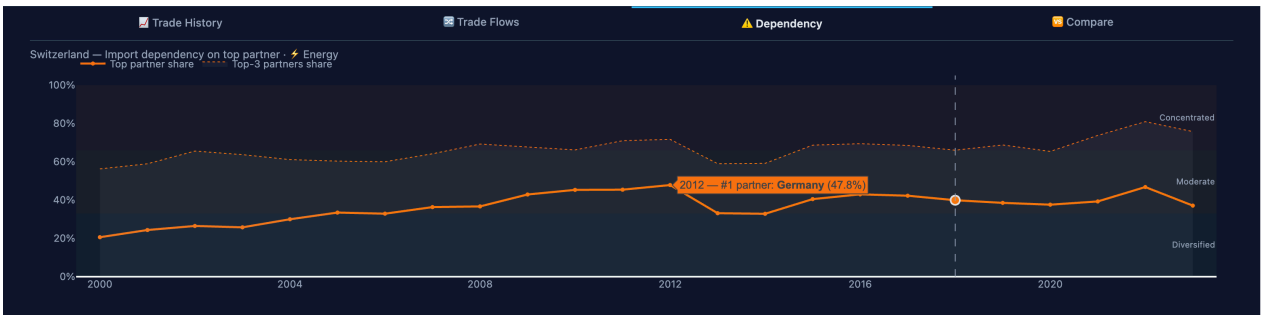

Figure 4: Dependency tab — Switzerland's energy import dependency on top partner

(E) Top Movers — year-over-year change

Global YoY % change (diverging bars, green gainers / red losers), filtered to countries above a trade-volume floor so rankings are not dominated by noise.

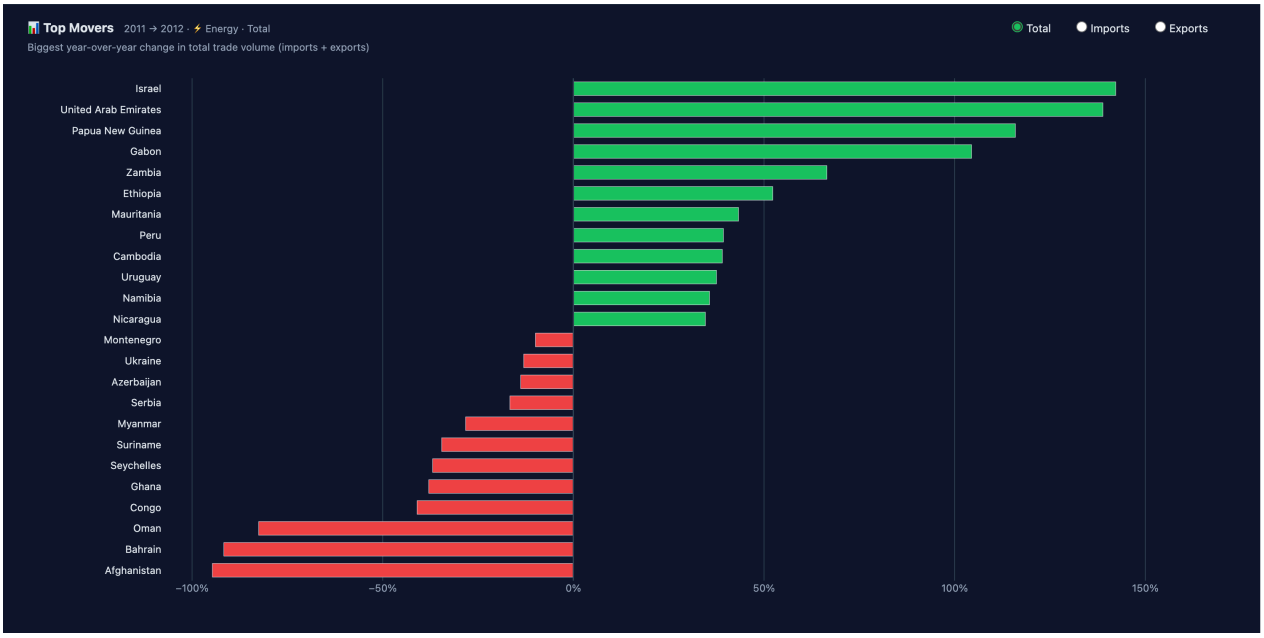

Figure 5: Top Movers chart

(F) Dependency Race — animated ranking

Animated ranking of "% of imports from top single partner" across years, revealing diversification or lock-in trends per commodity.

(G) Compare Mode — two countries side-by-side

Parallel mini-dashboards to contrast two countries (e.g., Germany vs Japan on steel) on totals, history, and dependency profile.

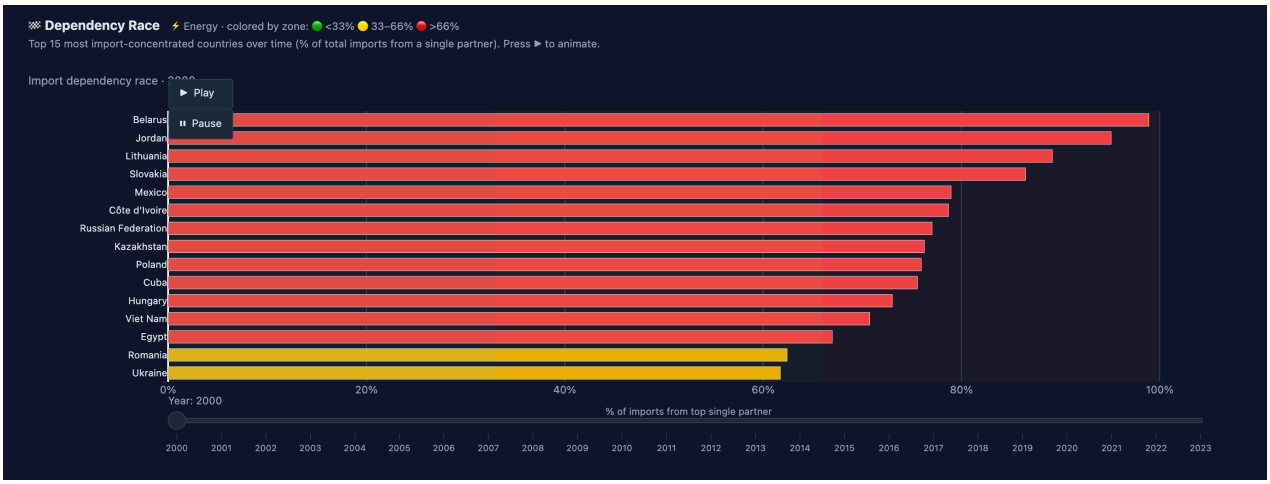

Figure 6: Dependency Race — animated bar-chart race

Figure 7: Compare tab — two-country side-by-side

4. MVP vs Extensions

Core (MVP — must ship for milestone 3)

1. Choropleth map + year slider + import/export toggle
2. Click-to-drill country detail panel (top-10 partners + history line)
3. Commodity dropdown covering ≥ 3 commodities
4. Sankey diagram of top bilateral flows
5. Dependency bar (top-5 partners + HHI) per country
6. Public deployment with the above reliably interactive

Each piece is **independent**: (1)–(3) rely only on `country_summary.csv`; (4)–(5) rely only on `partner_flow.csv`. They share no state other than the current selection, so we can ship them in any order.

Extensions (nice-to-have, droppable without breaking the narrative)

- **Disruption simulation** — user removes a country/link; the dashboard recomputes who loses what share of supply (propagation via re-routing).
- **Chord diagram** alternative view of the network (D3 embed).
- **Top Movers** and **Dependency Race** (already scaffolded).
- **"Explain this chart"** narrative side-captions (Storytelling lecture).
- **Search-as-you-type** country finder.
- **Mobile-friendly responsive layout.**

5. Tools & Lectures

Need	Tool	Lecture(s)
Map projection	Plotly `choropleth`, ISO-3	Maps; Perception
Time series	Plotly lines + shock bands	Time-series; Annotation
Flows	Plotly `sankey`	Graphs & networks
Rankings / composition	Plotly `bar`	Rankings; Composition
Animation	Plotly `frames`	Animation; Interaction
Layout & callbacks	**Dash** (Python, Flask)	Interaction; Web viz tooling
Data wrangling	**pandas**, **NumPy**	(prerequisite)
Deployment	Render / HF Spaces	(engineering)

Future lectures we expect to lean on: **Storytelling** (narrative captions), **Graphs & Networks** (Sankey refinement, optional force layout), **Evaluation & Perception** (color scales, legend validation).

6. Prototype Status & Risks

Running locally (`python app.py`, localhost:8050): choropleth, year slider, commodity dropdown, country detail panel with Trade History / Sankey / Dependency / Compare tabs, global Top Movers chart, Dependency Race animation, preprocessing pipeline, EDA notebook.

Gaps before milestone 3. Finish machinery & vehicles downloads; ship the disruption simulator; polish color scale and mobile layout; deploy a public URL.

Risks. *API quota* (500 calls/day) — mitigated by per-commodity CSV caching. *Missing ISO-3 codes* (~10% of partners) — manual override in `src/utils.py`. *Re-export inflation* — flagged in the EDA; dependency metric uses direct partner share. *Scope creep* — the MVP (§4) is independent of every extension, so extensions can be dropped without breaking the story.